READING PASSAGE 1

You should spend about 20 minutes on Questions 1–13, which are based on Reading Passage 1 below.

A Short History of Glass

A Long before people learned to make it, they used glass that nature had made. Obsidian, a black glass that forms when lava cools very quickly, was chipped into knives and arrowheads more than ten thousand years ago. Because it breaks into pieces with extremely sharp edges, it was valued so highly that it was carried and traded hundreds of kilometres from the volcanoes where it was found.

B The first glass made by people appeared in Mesopotamia around 3500 BC, probably as a by-product of metalworking or pottery. For many centuries it was used only for small objects such as beads and the shiny coating on clay pots. Making a vessel was slow and costly: workers wrapped hot glass around a core of sand and clay, and then scraped the core out once the glass had cooled.

C Everything changed in the first century BC, when craftsmen on the eastern coast of the Mediterranean discovered that glass could be blown. By blowing air through a metal pipe into a small ball of molten glass, a worker could form a thin bottle in minutes. Glass became cheap enough for ordinary households, and the Romans spread the technique to every corner of their empire.

D In medieval Europe, glassmakers were both admired and controlled. In Venice the industry was moved to the island of Murano in 1291, officially to protect the city from fire. The move also kept the secrets of the trade in one place: skilled workers were treated as important citizens, but they were not allowed to leave the republic.

E Clear window glass remained a luxury until the nineteenth century. Early panes were made by spinning a disc of glass, which left them thick in the middle and uneven at the edges. In 1959, the British engineer Alastair Pilkington introduced the float process, in which molten glass is poured onto a bath of liquid tin. The result is a perfectly flat sheet, and almost all window glass is still made this way today.

Questions 1–5

Reading Passage 1 has five paragraphs, A–E.
Choose the correct heading for each paragraph from the list of headings below.
Write the correct number, i–vii, in boxes 1–5 on your answer sheet.

List of Headings

i	A secret kept on an island
ii	A material shaped by volcanoes
iii	The search for a flat surface
iv	Glass in the computer age
v	A slow and expensive beginning
vi	A breath that changed an industry
vii	Why glass breaks

1 Paragraph A

2 Paragraph B

3 Paragraph C

4 Paragraph D

5 Paragraph E

Questions 6–9

Do the following statements agree with the information given in Reading Passage 1?

TRUE	if the statement agrees with the information
FALSE	if the statement contradicts the information
NOT GIVEN	if there is no information on this

6 Obsidian was used to make tools.

7 Glass vessels were common in Mesopotamia in 3500 BC.

8 The Romans learned glass blowing from the Venetians.

9 Pilkington's process uses a bath of liquid tin.

Questions 10–13

Complete the notes below.
Choose ONE WORD ONLY from the passage for each answer.

Early glass

- first used for beads and the coating on clay 10 …….
- vessels made around a core of sand and 11 …….

Venice

- industry moved to Murano to protect the city from 12 …….
- early panes made by spinning a 13 ……. of glass

READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

Why We Sleep

For most of history, sleep was seen as a time when the brain simply switched off. That view began to change in 1953, when Eugene Aserinsky, a student in Chicago, noticed that the eyes of sleeping people moved quickly behind their closed lids. These periods of rapid eye movement, or REM sleep, turned out to be the time when most dreaming happens, and they showed that the sleeping brain is far from idle.

Later research has linked sleep to memory. The psychologist Matthew Walker has shown that people who sleep after learning a list of words remember up to forty per cent more than people who stay awake. According to Walker, the brain uses deep sleep to move new information from a short-term store into long-term memory, rather like saving a document before closing the computer.

Sleep also seems to clean the brain. In 2013 the neuroscientist Maiken Nedergaard found that the spaces between brain cells grow larger during sleep, allowing fluid to wash away waste products. Some of these waste products are connected with illnesses that affect memory in old age, which suggests that poor sleep over many years may have serious effects.

Teenagers face a particular problem. Their internal clock shifts later during adolescence, so many of them are not tired until eleven o’clock at night or later. Schools that start at eight in the morning therefore force them to wake in the middle of their natural night. Several studies in the United States found that when schools moved their start time later, attendance improved and fewer students fell asleep in class.

Despite this evidence, adults in many countries now sleep less than their grandparents did. Artificial light, long working hours and the glow of phone screens all push bedtime later. Doctors recommend between seven and nine hours a night for adults, yet surveys suggest that about a third regularly get fewer than six.

Questions 14–17

Look at the following findings (Questions 14–17) and the list of researchers below.
Match each finding with the correct researcher, A, B or C.
Write the correct letter, A, B or C, in boxes 14–17 on your answer sheet.

14 Sleep helps the brain to remove waste.

15 Rapid eye movements happen during sleep.

16 Sleeping after studying improves memory.

17 Brain cells move apart during sleep.

List of Researchers

A	Eugene Aserinsky
B	Matthew Walker
C	Maiken Nedergaard

Questions 18–21

Complete each sentence with the correct ending, A–F, below.

A	when they start school later in the morning.
B	before they go to sleep.
C	in the middle of their natural night.
D	because they work longer hours.
E	during periods of REM sleep.
F	when they read on paper.

18 Most dreaming takes place

19 Teenagers who start school at eight wake up

20 Students attend school more often

21 Many adults sleep less than their grandparents

Questions 22–26

Answer the questions below.
Choose NO MORE THAN TWO WORDS AND/OR A NUMBER from the passage for each answer.

22 In which city did Aserinsky study?

23 By how much can sleeping after learning improve memory?

24 In which year did Nedergaard publish her finding?

25 What liquid washes waste out of the brain?

26 What fraction of adults often sleep less than six hours?

READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 below.

The Case for City Trees

City planners have long treated trees as decoration: pleasant to look at, but the first thing to be cut when a budget is reduced. I believe this attitude is a serious mistake. A growing body of research shows that trees are among the cheapest and most effective tools a city has for protecting the health of its people.

The clearest benefit is shade. On a summer afternoon, a street lined with mature trees can be up to ten degrees cooler than a bare one. As heatwaves become more frequent, this difference is not a matter of comfort but of survival, especially for older residents. Air conditioning can cool a single building, but only trees can cool an entire neighbourhood without using electricity.

Trees also reduce flooding. Their roots allow rainwater to soak into the ground instead of running across concrete into drains, and their leaves catch a surprising amount of rain before it reaches the ground. Some engineers argue that new drainage pipes are a better investment, but pipes must be replaced every few decades, while a well-chosen tree can live for a century or more.

The effect on mental health is harder to measure, yet it may be the most important of all. Studies in several countries have found that people who can see trees from their windows report lower levels of stress. Hospital patients with a view of greenery have even been reported to recover more quickly, although I suspect the size of this effect has sometimes been exaggerated.

None of this means that any tree in any place will do. A species that needs a great deal of water may die in a dry summer, and roots planted too close to buildings can damage foundations. Cities therefore need trained specialists who can choose the right tree for each street. Such experts are rare, and in my view training more of them should be a priority for every local council.

Questions 27–31

Do the following statements agree with the claims of the writer in Reading Passage 3?

YES if the statement agrees with the claims of the writer

NO if the statement contradicts the claims of the writer

NOT GIVEN if it is impossible to say what the writer thinks about this

27 City trees are often removed when money is short.

28 Air conditioning is a better way of cooling a neighbourhood.

29 Drainage pipes last longer than most trees.

30 The benefit of trees for hospital patients may have been overstated.

31 Most city councils already employ enough tree specialists.

Questions 32–35

Choose the correct letter, A, B, C or D.

32 According to the writer, why is shade important?

- **A** it makes streets more attractive
- **B** it can protect people's lives during heatwaves
- **C** it reduces the cost of electricity for the city
- **D** it helps trees to grow more quickly

33 How do trees reduce flooding?

- **A** by storing water in their trunks
- **B** by blocking drains with leaves
- **C** by helping rain soak into the ground
- **D** by raising the level of the street

34 What do some engineers prefer to trees?

- **A** larger parks
- **B** new drainage pipes
- **C** concrete walls
- **D** green roofs

35 What is the writer's main purpose in the passage?

- **A** to describe how trees grow in cities
- **B** to compare trees in different countries
- **C** to persuade readers that cities should invest in trees
- **D** to explain why trees damage buildings

Questions 36–40

Complete the summary using the list of words, A–H, below.

Choosing city trees

A tree that needs a lot of 36 …… may not survive a dry summer, and trees planted too near houses can harm their 37 …… For this reason cities need 38 …… who can select suitable trees. The writer believes that 39 …… councils should make the 40 …… of these people a priority.

A water B foundations C specialists D local

E training F sunlight G windows H national

ANSWERS

1 ii	2 v	3 vi	4 i
5 iii	6 TRUE	7 FALSE	8 NOT GIVEN
9 TRUE	10 pots	11 clay	12 fire
13 disc	14 C	15 A	16 B
17 C	18 E	19 C	20 A
21 D	22 Chicago	23 40%|forty per cent	24 2013
25 fluid	26 a third|one third	27 YES	28 NO
29 NO	30 YES	31 NO	32 B
33 C	34 B	35 C	36 A
37 B	38 C	39 D	40 E